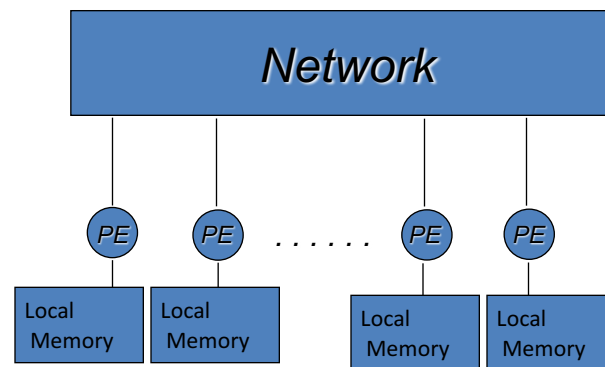


Target Environment: MIMD.
Multiple Tasks Per Processor.



1

Assumptions

- System assumed MIMD.
- Multiple jobs in system at any time.
- Jobs assumed parallel or sequential.
- Local Scheduling at each processor.
- Each Processor capable of self monitoring its performance.

2

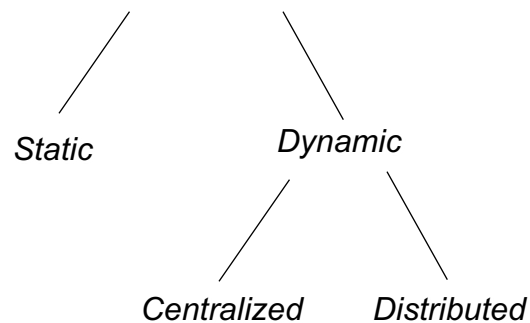
Load Balancing: Definition 1

Load Balancing is the activity of migrating load units from one processing element (PE) to another so that the number of load units in different PEs is *approximately* the same.

3

Load Balancing: Classification

Load Balancing



4

Load Balancing: Classification

- **Static:**
 - Task information (execution time) is computed from the application before load distribution. The allocation remains the same throughout the application's execution
- **Dynamic:**
 - No *a priori* task information is used in initial distribution. Must satisfy changing requirements by making task distribution decisions at run-time

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Load Balancing: Classification

- **Centralized:**
 - Load Balancing decisions are made by one processing element only, which is responsible for maintaining global load information.
- **Distributed:**
 - Load Balancing decisions are made locally. Global load information is distributed among all processing elements and they all share the responsibility of achieving global load balance.

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Load Balancing: Classification

- Implicit:
 - Load Balancing performed automatically by the system
- Explicit:
 - User decides which tasks should migrate and when

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Load Balancing: Phases

- When to load balance
- Where to migrate the tasks
- How many tasks to transfer/request
- Which tasks to transfer

8

Load Balancing: Advantages

- Improves single application overall performance by redistributing the workload among processors.
- Increases throughput if multi-application environment.
- System function, hence transparent to user.

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Load Balancing: Disadvantages

- Load Balancing activity comes at the expense of useful computation.
- Incurs into communications overhead:
 - Transfer of tasks among Pes.
 - Cost of collecting load information.
 - Message forwarding.
- Space required to maintain load balancing information.

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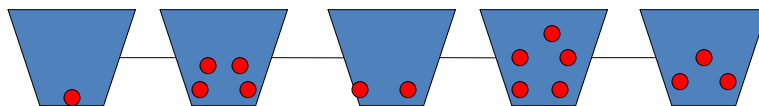
Load Balancing: Previous work

- Gradient Model
- Central Job Dispatcher
- Sender Initiated/Receiver Initiated Diffusion
- Randomize seeking
- Dimension Exchange Method
- Hierarchical Balancing Method

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According to Definition 1

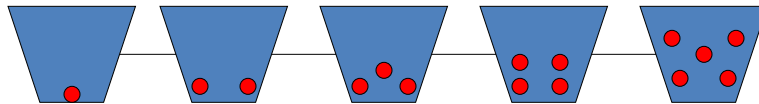
- Fully distributed algorithm:



- Can look at neighbors only and transfer marbles so that two neighbors have same number of marbles + or - 1
- Converges to following configuration:

12

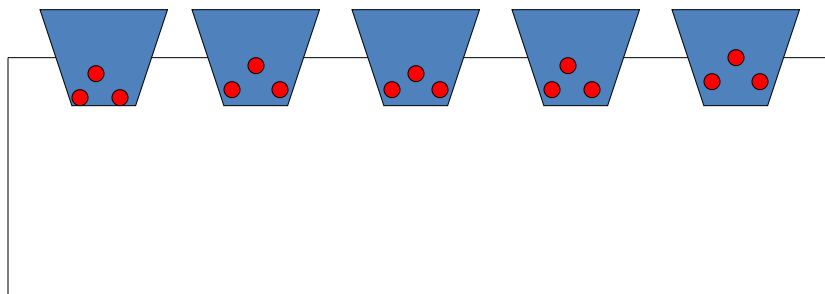
According to Definition 1



- Is this BALANCED ?
- Solution ?

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According to Definition 1



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Load Balancing: Definition 2

Load Balancing is the activity of migrating load units from one processing element (PE) to another so that PEs have at least one load unit at all times (if enough workload is available).

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Load Balancing: Problem Statement

Redistribute workload among processors to increase throughput by maximizing processor utilization.

Minimize processor idle time with minimum task redistribution overhead.

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Load Balancing: Goals

- To minimize processor idling time.
- To minimize overhead:
 - communications
 - load balancing activity
- To be stable.
- To quickly (re-)distribute the load.

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Rate-of-Change Load Balancing

- The load balancing algorithm proposed for:
 - High throughput
 - High Processor utilization
 - Fault tolerant
 - On-line repairable
 - Scalable

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Rate of Change Load Balancing (RoC-LB)

- Dynamically compute the way in which local load changes in time (rate of change).
- Redistribute load *if and only if future starvation is predicted*.

– Assumes time slicing at each PE (time interval).

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RoC-LB: Our Approach

- Uses local tables containing system's load distribution information to determine where tasks should be transferred to.
- Uses three thresholds to determine when and how many tasks to transfer.
- Uses task size and previous execution time to determine which tasks to transfer.

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RoC-LB: Phases

- When to load balance
- Where to migrate the tasks
- How many tasks to transfer/request
- Which tasks to transfer

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RoC-LB: When Phase

Each Processor

- Calculates its local rate of change (DL).
- Uses DL as predictor of the probable number of tasks in subsequent intervals.
- If DL is negative, computes number of sampling intervals before idle state reached.

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RoC-LB: When Phase (cont'd)

Each Processor

- Initiates a migration request if number of intervals to starvation is less than network delay (ND).

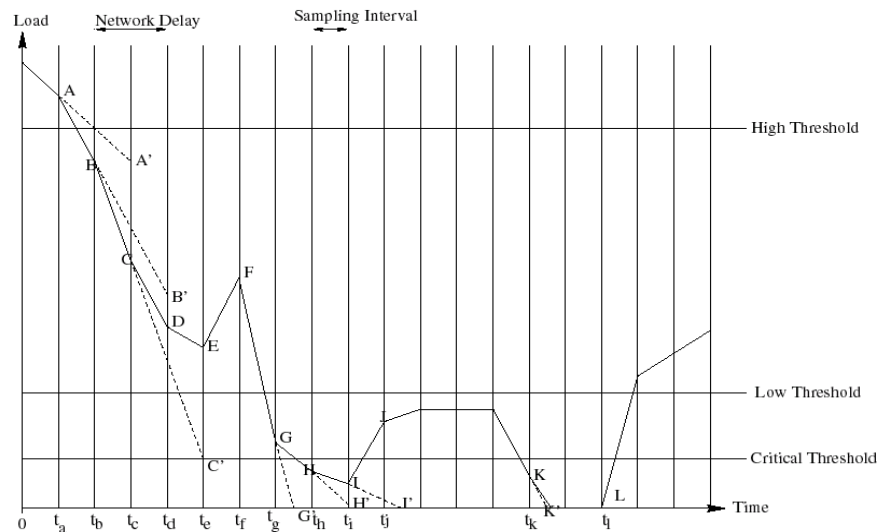
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RoC-LB: When Phase (cont'd)

- Two thresholds are used to determine the load status of a PE:
 - Above HT : Source PE
 - Below LT : Sink PE
 - Between HT and LT : Neutral PE
- Two exceptions:
 - When the load falls below CT
 - When the load is above HT

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RoC-LB depicted in time



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RoC-LB: Where Phase

- Sink processor initiates process
 - Select source processor from table
 - Sends request to selected processor
 - Deletes source entry from table
 - Upon receiving reply updates table

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RoC-LB: Where Phase

- Receiving processor
 - Transfers tasks to sink if still source
 - Increments count
 - Forward request to another source processor if no longer source or not able to satisfy whole request
 - Updates table based on request message

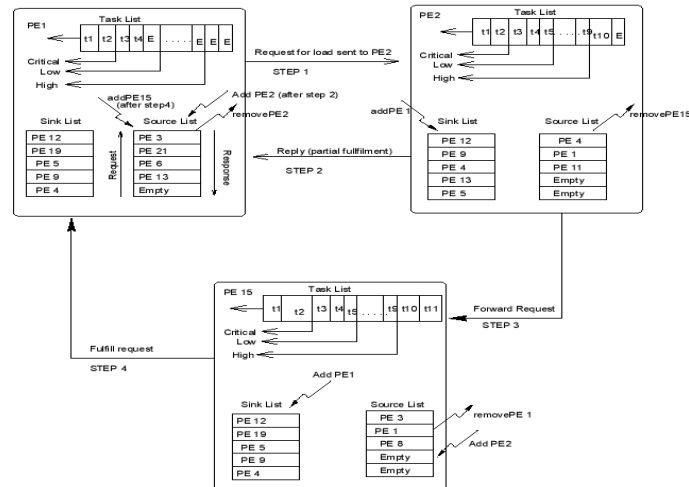
27

RoC-LB: Transfer request protocol

- Request Fields:
 - Sink processor identification
 - number of load units
 - load state of preceding processor
 - count number

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ROC-LB: Transfer request protocol



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RoC-LB: How many / Which Phases

- Sender Side: Assuming DL constant for the duration of a ND calculate Predicted Load (PL). Request $\text{abs}(\text{PL})$
- Receiver Side: Transfer tasks above HT
- Older Tasks. Higher probability of living long enough to amortize their migration cost

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ROC-LB: Classification

- Dynamic
- Distributed
- On demand
- Preemptive
- Implicit

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RoC-LB: Experiments

- Each simulation uses 32 000 tasks.
- Tasks have different sizes and life span.
- Four different Load Balancing Algorithms used.
- Four different network topologies:
 - mesh
 - hypercube
 - fully connected network
 - Network of Workstations

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RoC-LB: Performance Metrics

Normalized Performance

$$NP = \frac{T_{noLB} - T_{LB}}{T_{noLB} - T_{opt}}$$

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RoC-LB: Results

Symmetric distribution

	Mesh	Hypercube	Fully Connected	Network of workstations
Gradient Model	0.66	0.69	0.73	0.52
Sender Initiated Diffusion	0.70	0.71	0.74	0.59
Central Job Dispatcher	0.59	0.62	0.73	0.37
Rate of Change Model	0.79	0.79	0.87	0.61

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RoC-LB: Results

Asymmetric distribution

	Mesh	Hypercube	Fully Connected	Network of workstations
Gradient Model	0.72	0.74	0.76	0.54
Sender Initiated Diffusion	0.65	0.65	0.68	0.52
Central Job Dispatcher	0.65	0.66	0.65	0.45
Rate of Change Model	0.74	0.75	0.81	0.60

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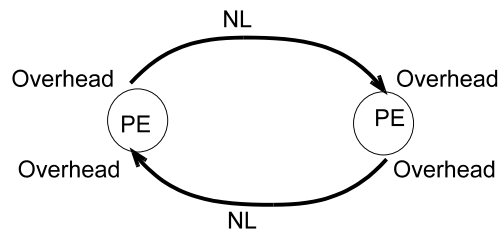
RoC-LB: Results

Symmetric distribution with Arrivals

	Mesh	Hypercube	Fully Connected	Network of workstations
Gradient Model	0.70	0.77	0.81	0.54
Sender Initiated Diffusion	0.73	0.76	0.81	0.58
Central Job Dispatcher	0.52	0.53	0.53	0.42
Rate of Change Model	0.82	0.82	0.88	0.62

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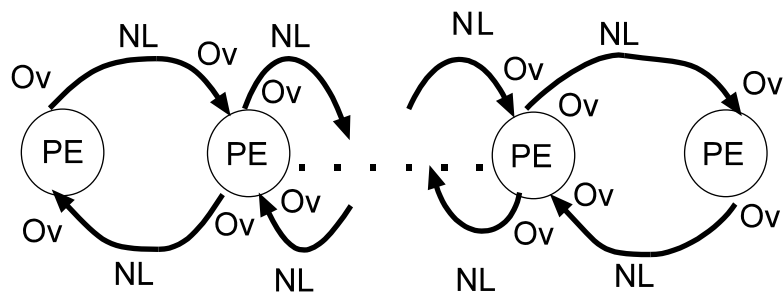
Influence of ND in RoC-LB



$$ND = 4 \times \text{Overhead} + 2 \times \text{Network Latency}$$

37

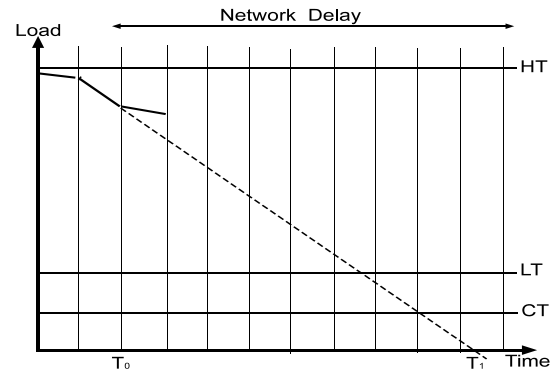
Influence of ND in RoC-LB



$$ND = \text{NumberHops} \times (2 \times \text{NL} + 4 \times \text{Ov})$$

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Influence of ND in RoC-LB



Very sensitive to even small changes in the load

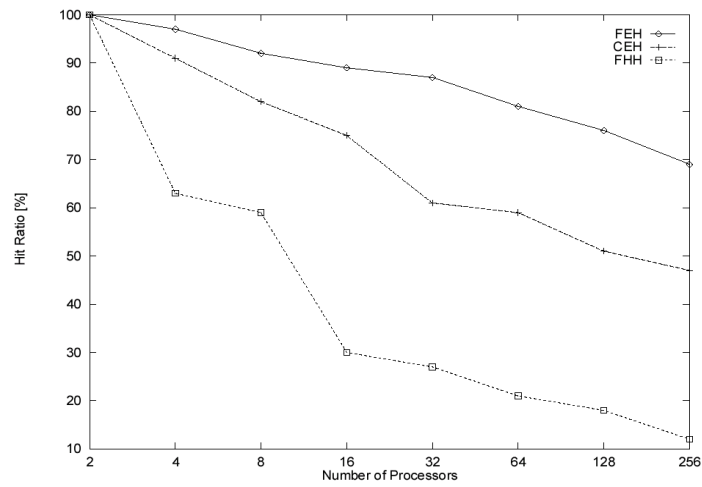
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RoC-LB: Table information accuracy

- First Hop Hit
 - The first entry in the table is accurate in the first hop
- First Entry Hit
 - The first entry in the table leads to a load transfer
- Complete Entry Hit
 - The load request is completely satisfied

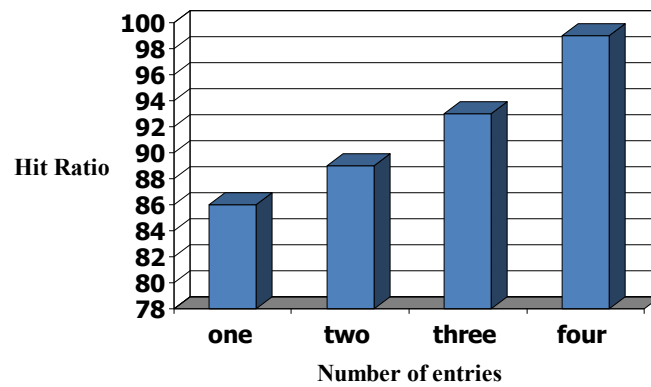
40

RoC-LB: Results (experiment 1)



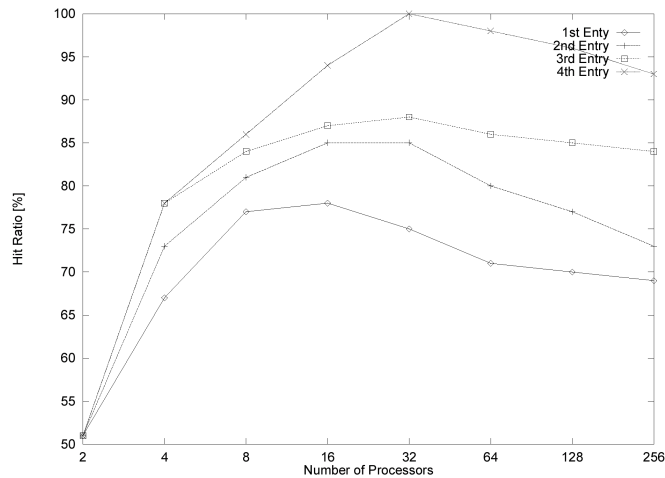
41

RoC-LB: Results (experiment 1)



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RoC-LB: Results (experiment 2)



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RoC-LB: Improvements

- How to increase the probability of a complete hit in shorter time?
 - Alternative 1: Every PE sends a load request to all entries in the source table.
 - Alternative 2: Only the request initiator PE sends a load request to all entries in the source table. Any intermediate PE forwards the message to just one other PE.

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RoC-LB: Improvements

- How to increase the probability of a complete hit in shorter time?
 - Alternative 3: The initiator PE multicasts a request to all PEs in its source table.
 - Each request will contain a different number of loads units or
 - Each request will be forwarded a different number of times.

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Conjecture

Parallel Scheduling > Local Scheduling +
Load Balancing

“For every Parallel Schedule, there exist an equally effective Local Scheduling + Load Balancing”

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